

# Intermediate Algebra Final Test

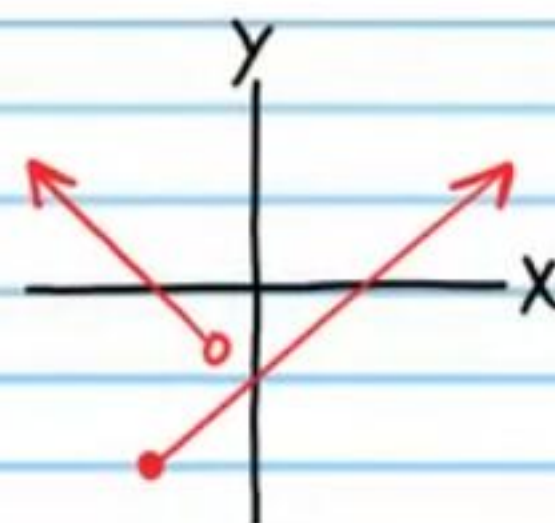
- I) This is a cumulative exam.
- II) Write your name, the date, and the course title (i.e. Intermediate Algebra).
- III) Show all work.
- IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- V) You will need a calculator for this test.
- VI) You do not have to write the Roman numeral for each exponent rule that you use on this test.
- VII) Do not leave your answers with fraction exponents, negative exponents, zero power exponents, or 1<sup>st</sup> power exponents.
- VIII) Each problem is worth eight points for a total of 200 points.

① Circle the graph or table that cannot represent a function.

| x  | y  |
|----|----|
| -2 | 3  |
| 0  | 0  |
| 2  | -1 |
| 5  | 4  |
| -2 | 5  |
| 5  | 4  |



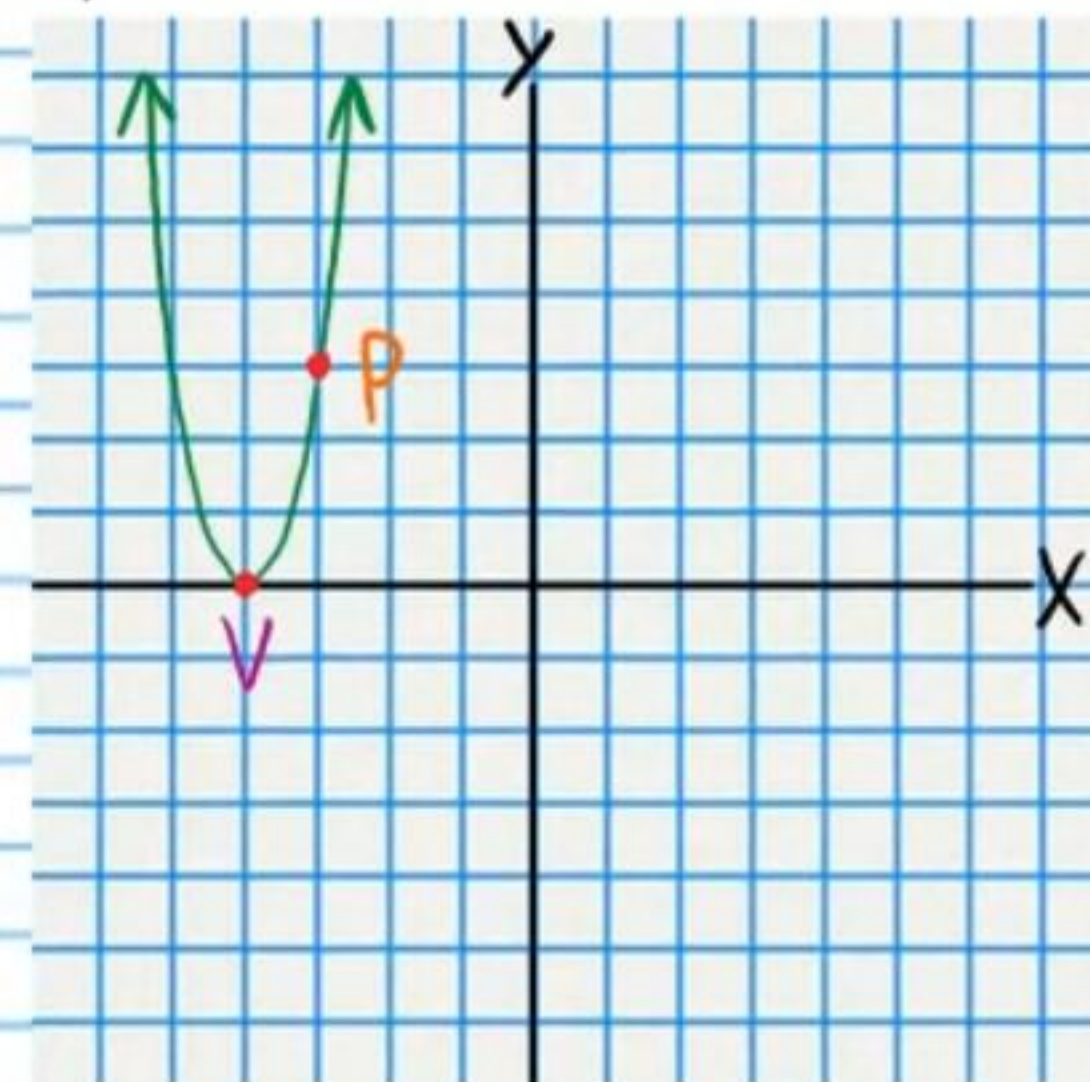
| x  | y  |
|----|----|
| -4 | 10 |
| -3 | 10 |
| 10 | -3 |
| -1 | 10 |
| 0  | 5  |
| -3 | 7  |



② Simplify.

$$(2x^{\frac{1}{2}} \cdot y^4)^2 (2xy \cdot y^{\frac{1}{2}})^3$$

③ i) Write an equation of the parabola. Hint: It is easiest to use vertex form.



ii) Find the discriminant of the equation below and use it to determine the number of x-intercepts of the corresponding graph.

$$y = -4x^2 + 4x - 1$$

④ Solve. Write exact expression(s) for your answer.

$$\log_6(5x-3) + \log_6 x = 1$$

⑤ Solve.

$$2\log x - 2\log 2 = 2$$

⑥ Solve. Round to the nearest hundredth.

$$-7 = 2 + 3\ln(2x-1)$$

⑦ Rationalize and simplify.

$$\frac{\sqrt{5} + \sqrt{3}}{\sqrt{2} + \sqrt{7}}$$

⑧ Solve.

$$\frac{1}{216^{-x}} = \sqrt{36^3}$$

⑨ Find the algebraic solution and the graphical solution.

$$-\frac{|-4x+4|}{5} + 7 \leq 6$$

⑩ i) Solve.

$$\log_8\left(\frac{9}{3^x}\right) = \log_8 \sqrt{3}$$

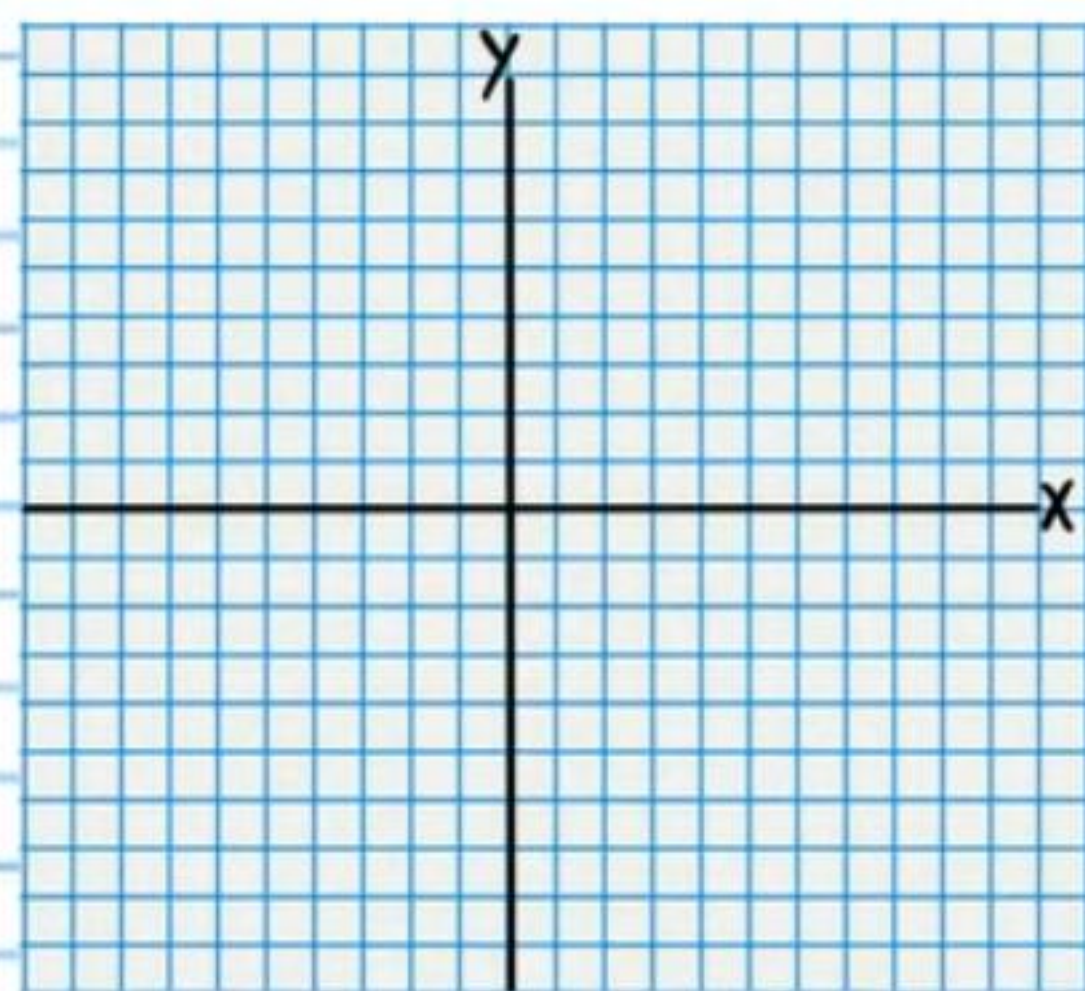
ii) Solve.

$$1 = \frac{8^{x-4}}{2^{x-4}}$$

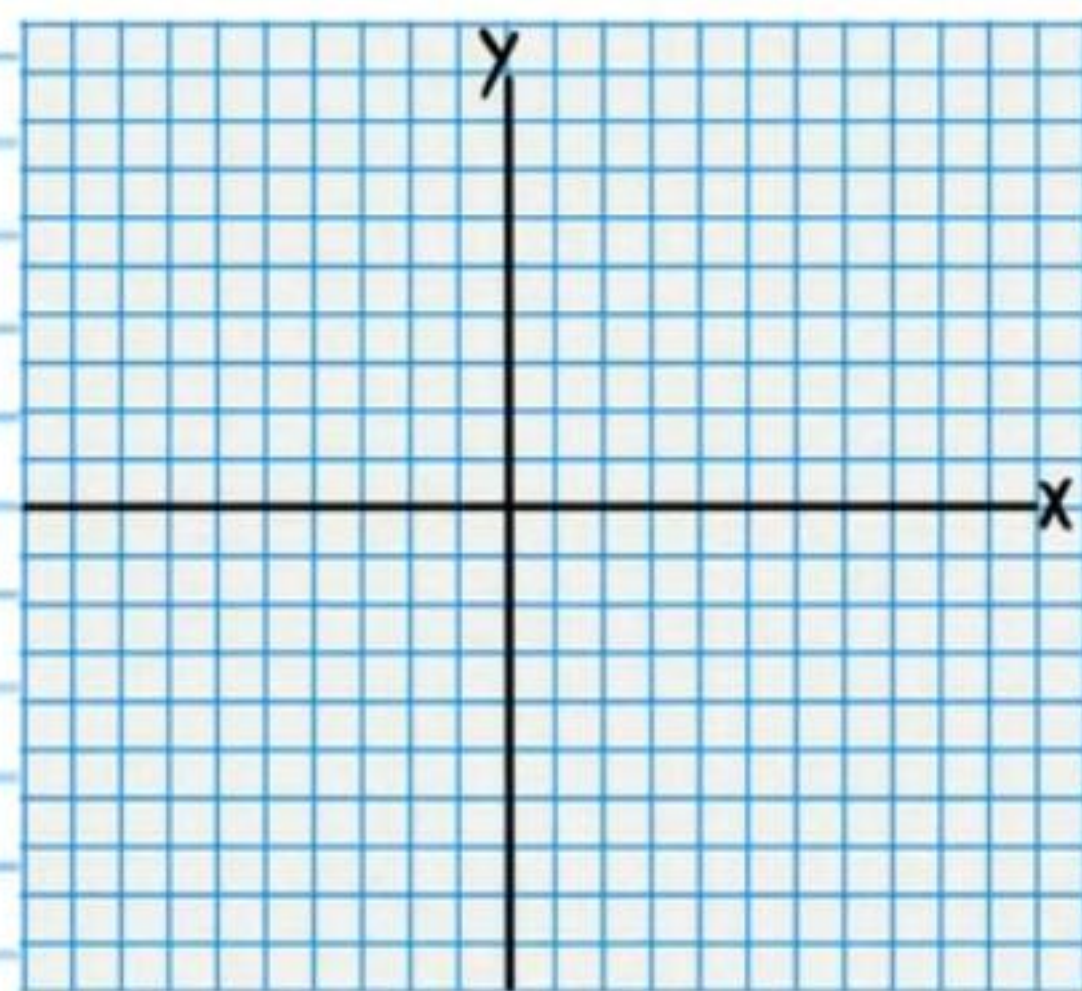
⑪ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

$$f(x) = \frac{4}{x}$$

Parent Function



Transformation



ii) Write the domain and range of the function in interval notation.

Domain:

Range:

⑫ If  $f(x) = (x+3)^2$  and  $g(x) = \frac{1}{x} + 3$ , Find the following:

i)  $(f \cdot g)(-1)$

ii)  $(f \circ g)(x)$

Find the values below if  $h(x) = \begin{cases} 2|x|, & -7 \leq x \leq -1 \\ 3-x, & 0 \leq x \leq 8 \end{cases}$

iii)  $h(0)$

iv)  $h(7\frac{1}{2})$

v)  $h(-6)$

vi)  $h(-8)$

vii) Write the domain of  $h(x)$  in interval notation.

viii) Write the domain of  $h(x)$  in set notation.

⑬ Write in  $a+bi$  form.

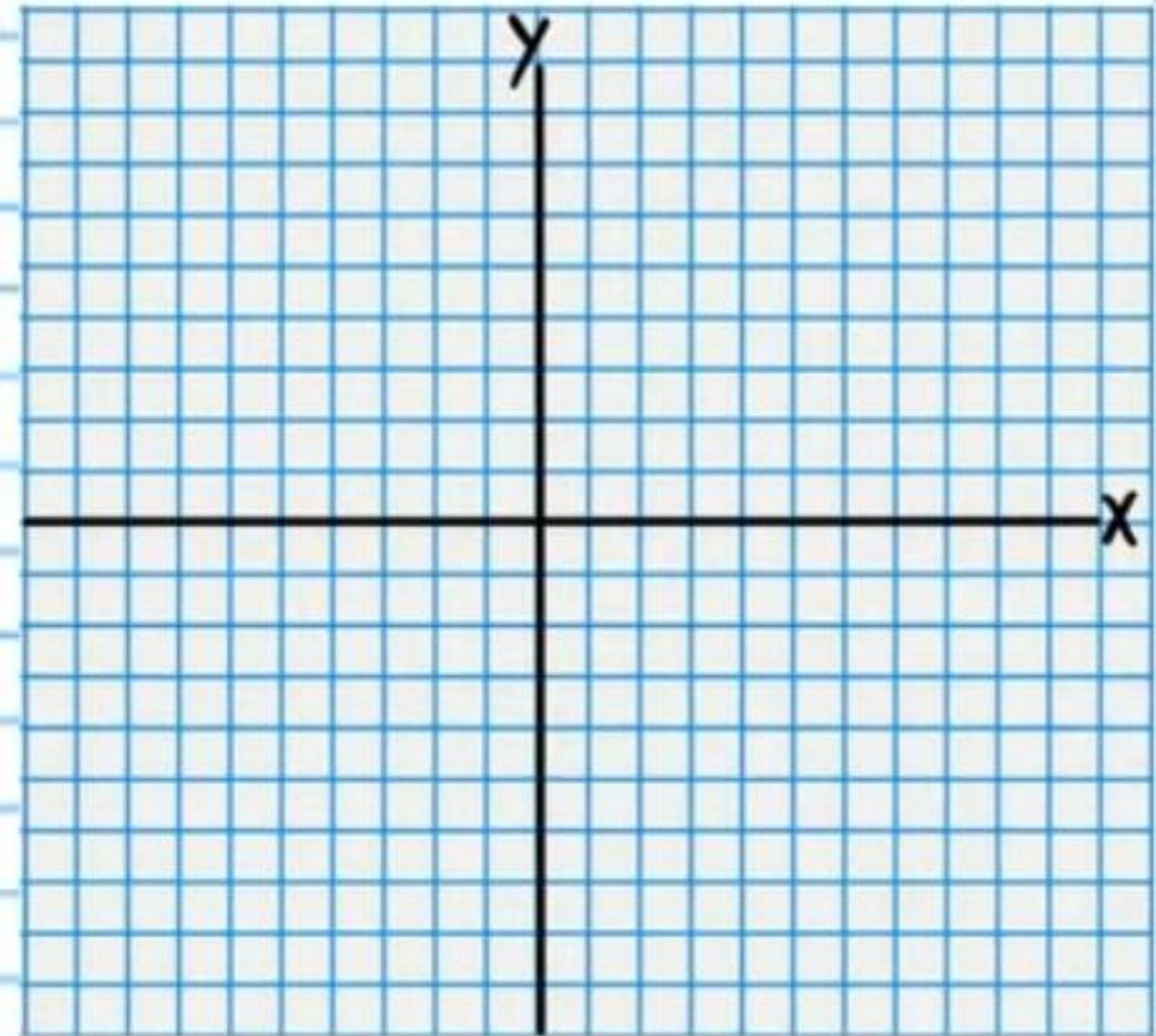
$$\sqrt{-80} + 2\sqrt{-125} + i^{42}$$

⑭ Convert to the other two forms (i.e. intercept and standard).

$$-(x+2)^2 + 1 = y$$

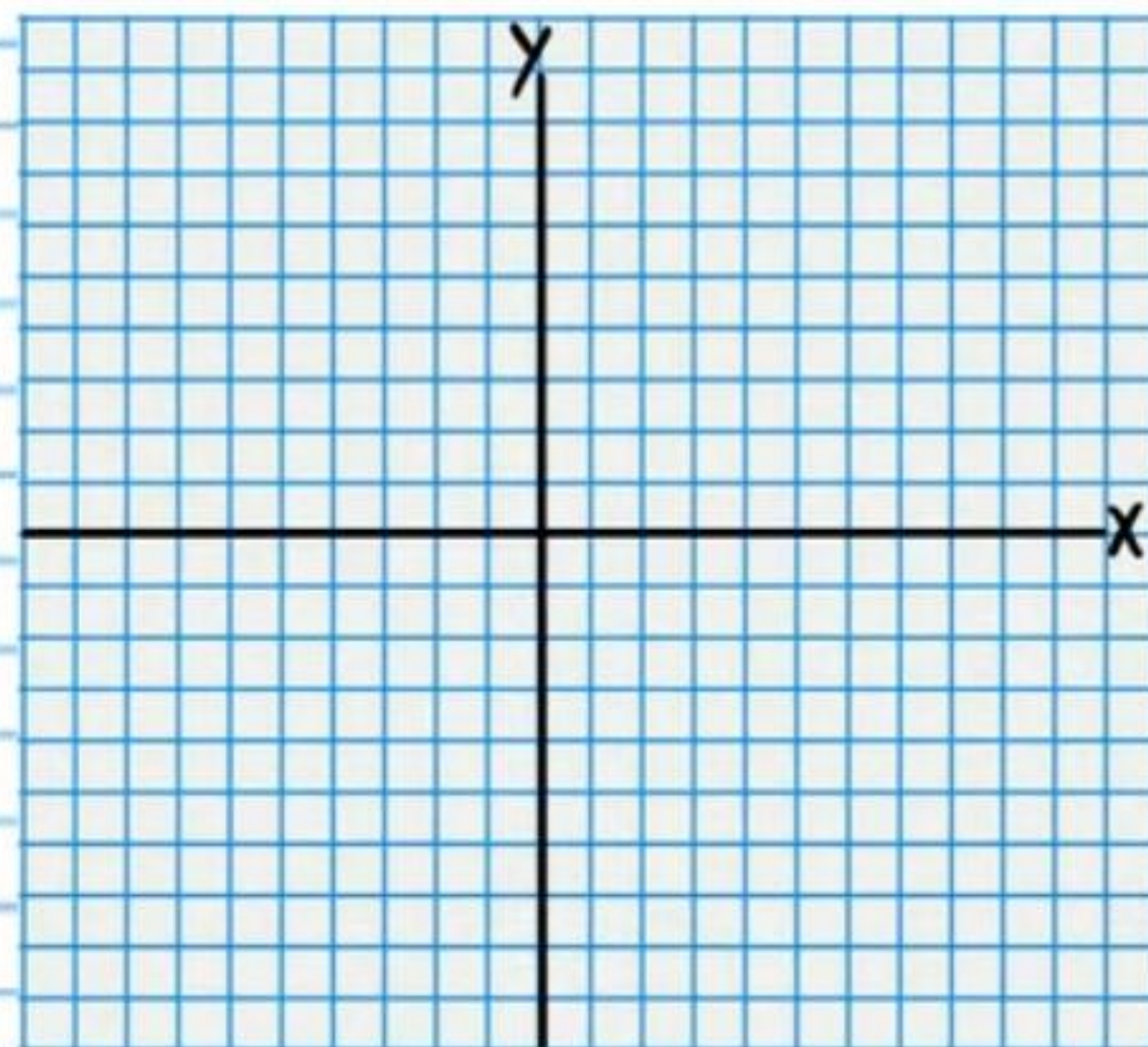
⑮ Graph the following quadratic expression. Draw the line of symmetry and write its equation. Write the coordinates of the vertex, x-intercepts, and y-intercept, and draw the points on the graph.

$$y = x^2 + 4x + 1$$



⑩ Graph the following quadratic expression. Write the equation of the directrix, the coordinates of the Vertex, and the Coordinates of the x-intercepts. Draw the directrix as a dotted line and write each point on the graph.

$$-\frac{1}{8}(x+9)(x+1)=y$$



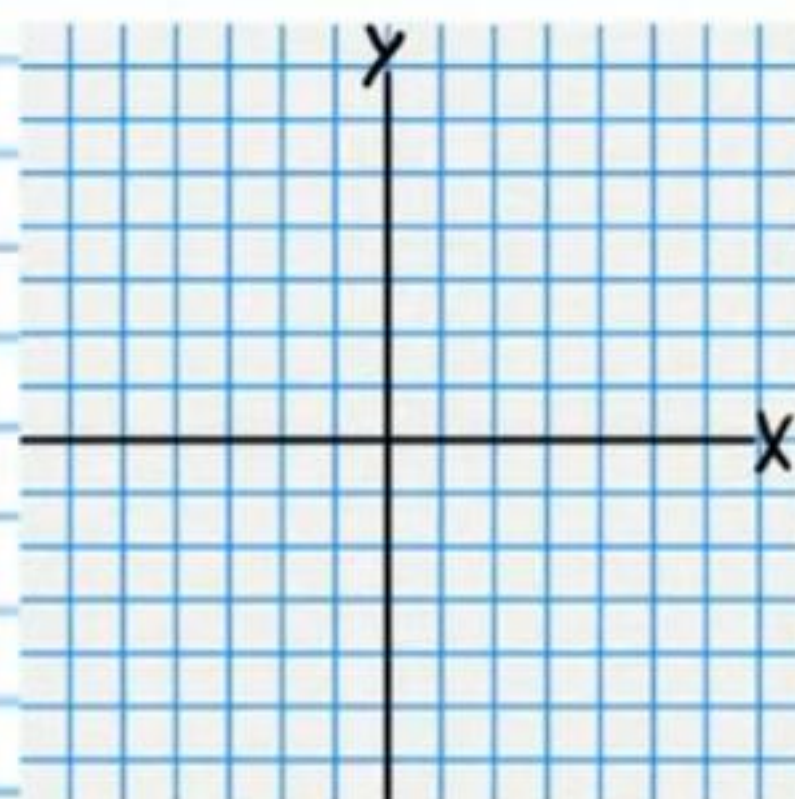
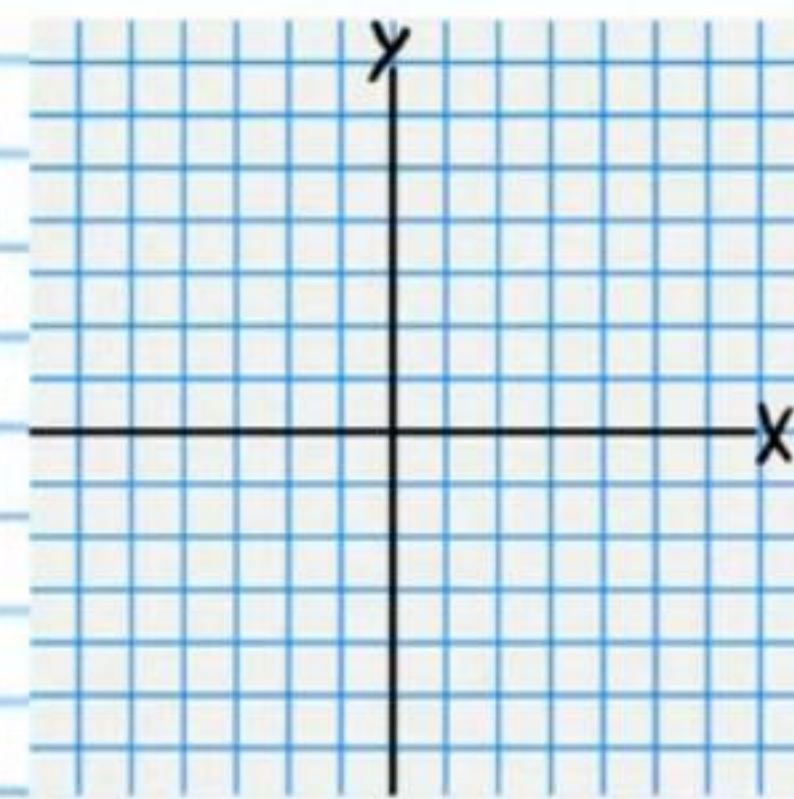
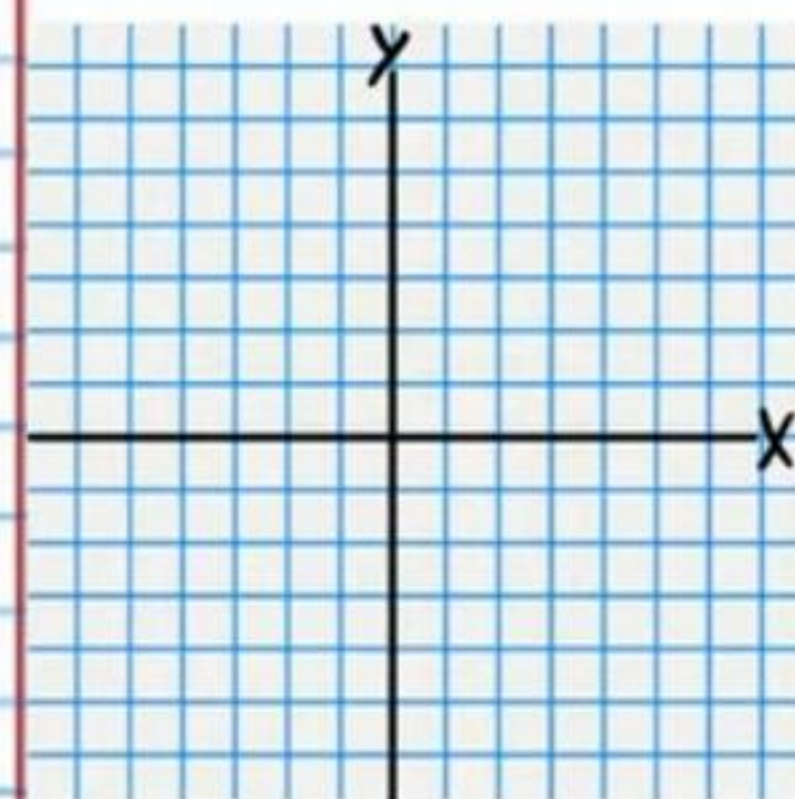
⑪ i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation.

$$f(x) = -\sqrt{x-3} + 4$$

Parent Function

Transformation #1

Final Transformation



ii) Write the domain and range of the function in set notation.

Domain:

Range:

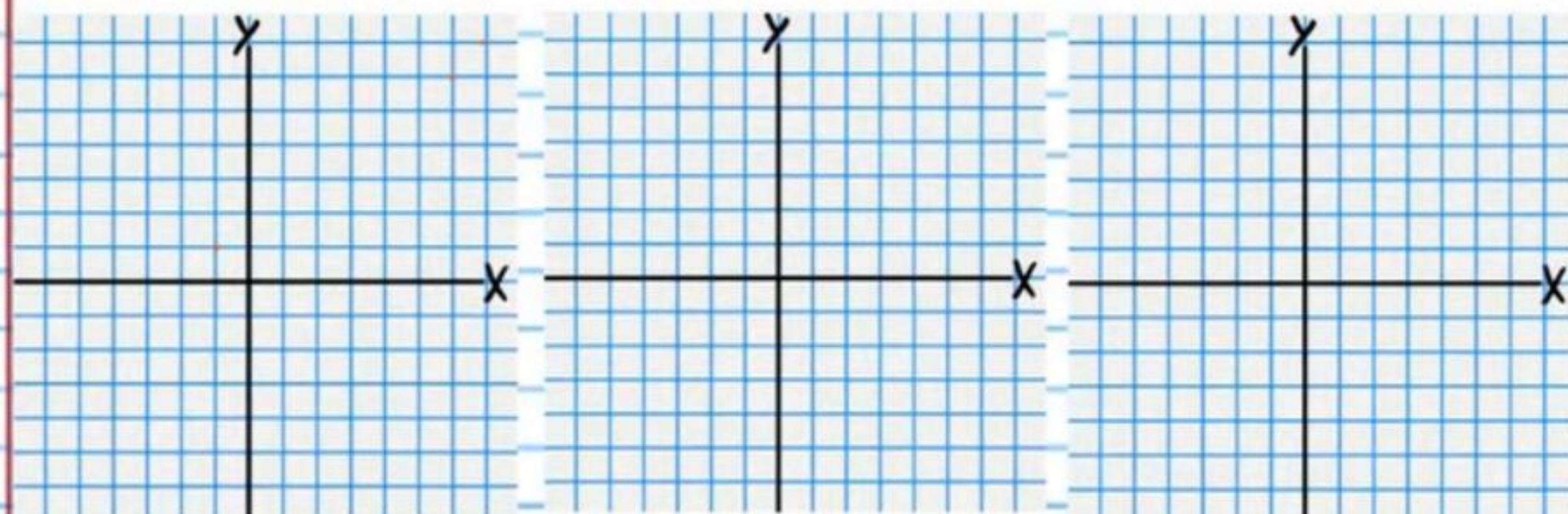
⑩ i) Graph the function below by performing multiple transformations on the graph of the parent function. *Graph each transformation and write its equation.*

$$f(x) = \left| \frac{x}{2} + 2 \right| + 3$$

Parent Function

Transformation #1

Final Transformation



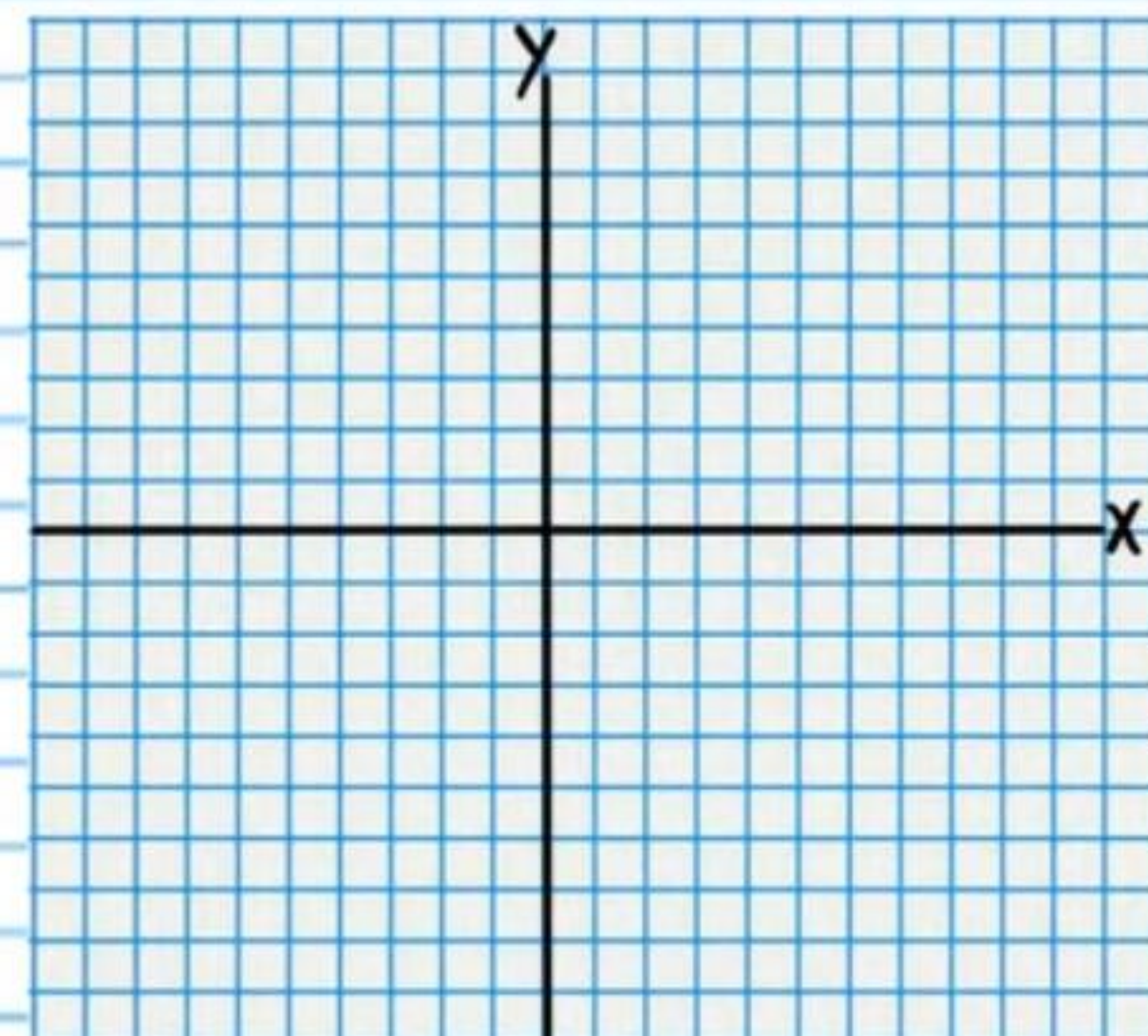
ii) Write the domain and range of the function in interval notation.

Domain:

Range:

⑪ Graph the following quadratic expression. Write the coordinates of the vertex, the x-intercepts, and the focus, and draw the points on the graph.

$$y = (x-1)^2 - 4$$



②① Rationalize and simplify.  
Assume that  $x$  is positive.

$$\sqrt[4]{\frac{x^{13}}{32}}$$

②① Solve by completing the square.

$$0 = x^2 - 14x + 24$$

②② i) Simplify.

Cube root.

Square root

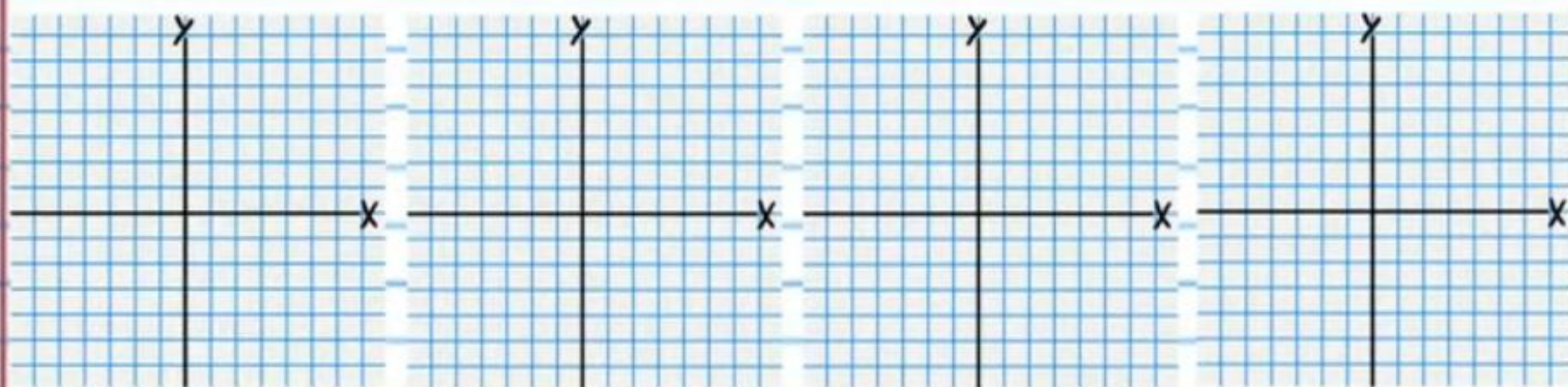
$$\sqrt[3]{8x^3} + 10\sqrt{56x^3} - 3x\sqrt{14x} + 8x$$

ii) If  $y$  is directly proportional to  $X$ , and  $y = 88$  when  $x = 4$ , find  $y$  when  $x = 3$ .

23) i) Graph the function below by performing multiple transformations on the graph of the parent function. Graph each transformation and write its equation. Draw the asymptote(s) as dotted lines.

$$f(x) = \frac{3^{-x+1}}{2} + 2$$

Parent Function    Transformation #1    Transformation #2    Final Transformation



ii) Write the domain and range of the function in set notation.

Domain:

Range:

24) i) If an investment grows from \$2800 to \$3612 at a 0.9% annual interest rate compounded continuously, how much time passed during this growth?

ii) If \$82,023 is deposited in a bank account paying a 1.1% annual interest rate compounded daily, how much money will be in the account in  $3\frac{1}{2}$  years? Assume that there are no withdrawals or additional deposits.

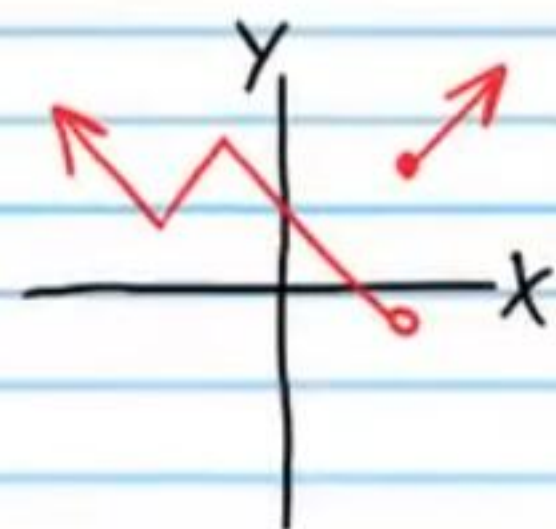
②⑤ Solve.

$$\begin{cases} 5x + 3y + 2z = -15 \\ -x + 2y - 3z = 16 \\ 2x - y + z = -11 \end{cases}$$

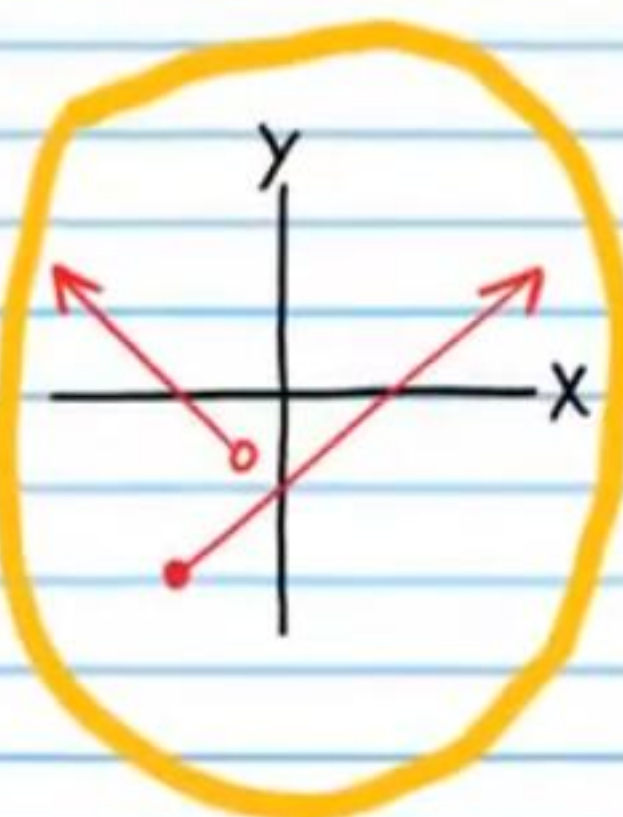
# Answers

①

| x  | y  |
|----|----|
| -2 | 3  |
| 0  | 0  |
| 2  | -1 |
| 5  | 4  |
| -2 | 5  |
| 5  | 4  |



| x  | y  |
|----|----|
| -4 | 10 |
| -3 | 10 |
| 10 | -3 |
| -1 | 10 |
| 0  | 5  |
| -3 | 7  |



②  $32x^4\sqrt{y^{25}}$

③ i)  $y = 3(x+4)^2$  ii)  $D=0$ , One x-intercept

④  $x = \frac{3+\sqrt{129}}{10}$  or  $x \neq \frac{3-\sqrt{129}}{10}$

⑤  $x = 20$  or  $x \neq -20$

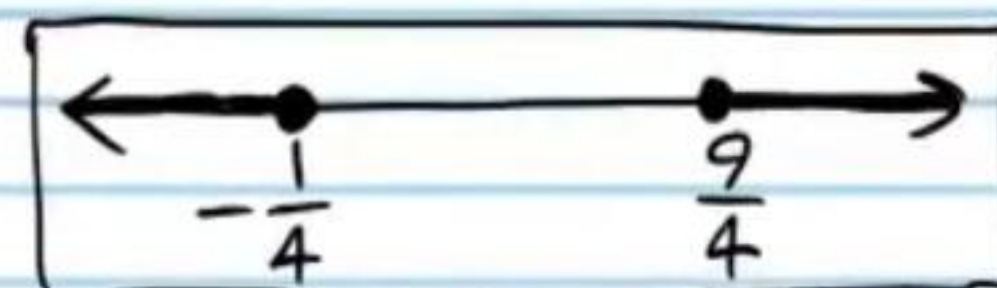
⑥  $x \approx 0.52$

Different forms

⑦  $\frac{\sqrt{10} - \sqrt{35} + \sqrt{6} - \sqrt{21}}{-5}$  or  $\frac{-\sqrt{10} + \sqrt{35} - \sqrt{6} + \sqrt{21}}{5}$

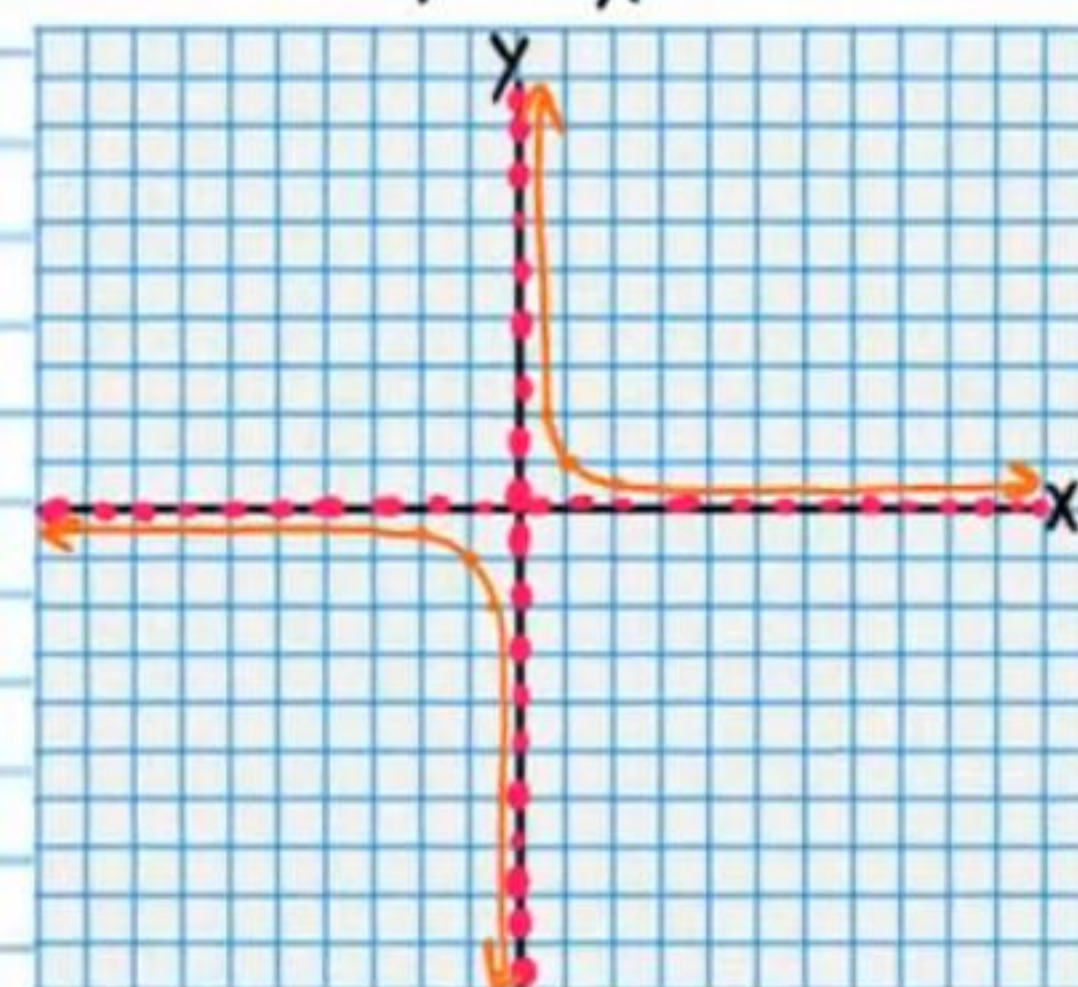
⑧  $x = 1$

⑨  $x \leq -\frac{1}{4}$  or  $x \geq \frac{9}{4}$

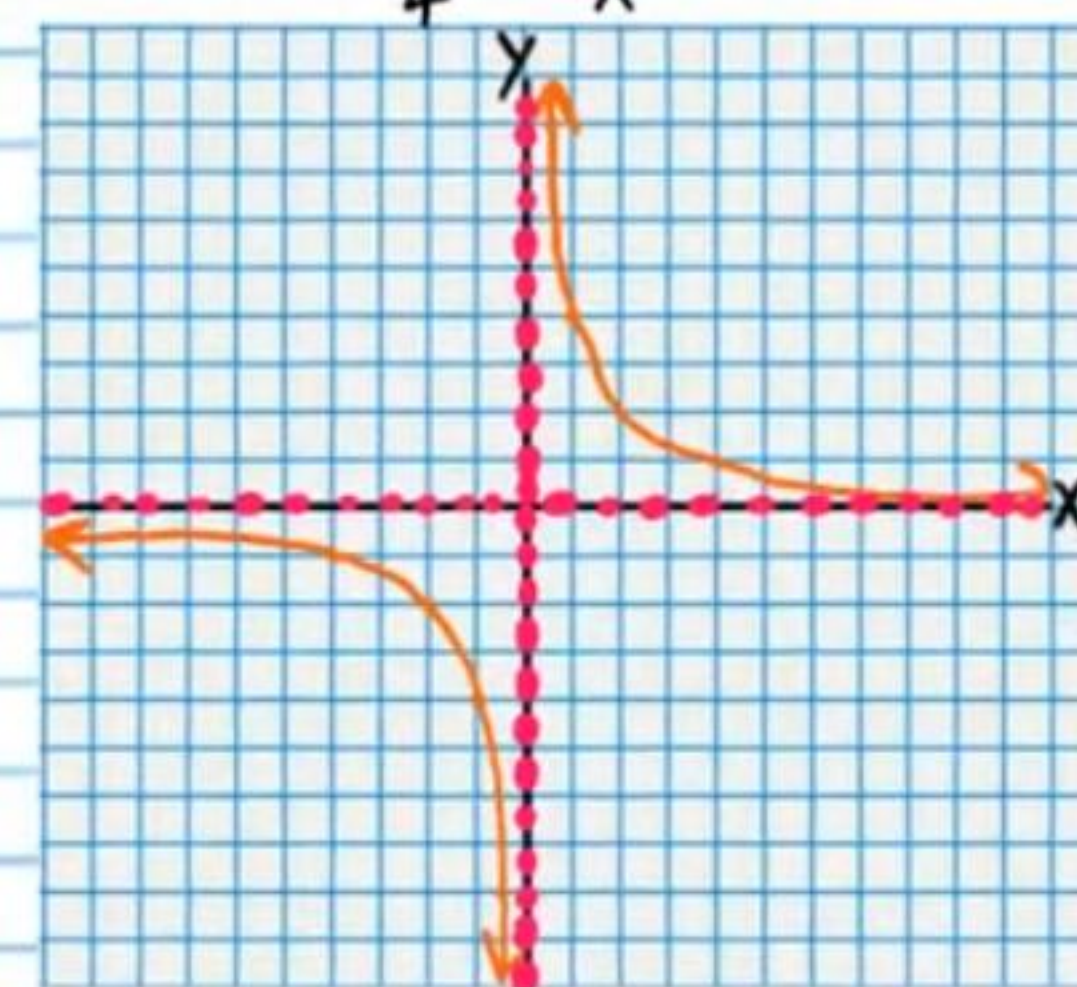


⑩ i)  $\frac{1}{2} = x$  ii)  $x = 4$

⑪ i) Parent Function  $y = \frac{1}{x}$



Transformation  $\frac{y}{4} = \frac{1}{x}$



ii) Domain:  $(-\infty, 0) \cup (0, \infty)$  Range:  $(-\infty, 0) \cup (0, \infty)$

⑫ i)  $8$  ii)  $(\frac{1}{x} + 6)^2$  iii)  $3$  iv)  $-4\frac{1}{2}$  or  $-4.5$

v)  $12$  vi) Undefined vii)  $[-7, -1] \cup [0, 8]$

viii)  $\{x \mid -7 \leq x \leq -1 \text{ or } 0 \leq x \leq 8\}$

⑬  $-1 + 14\sqrt{5}i$

⑭  $-x^2 - 4x - 3 = y$   $-(x+3)(x+1) = y$

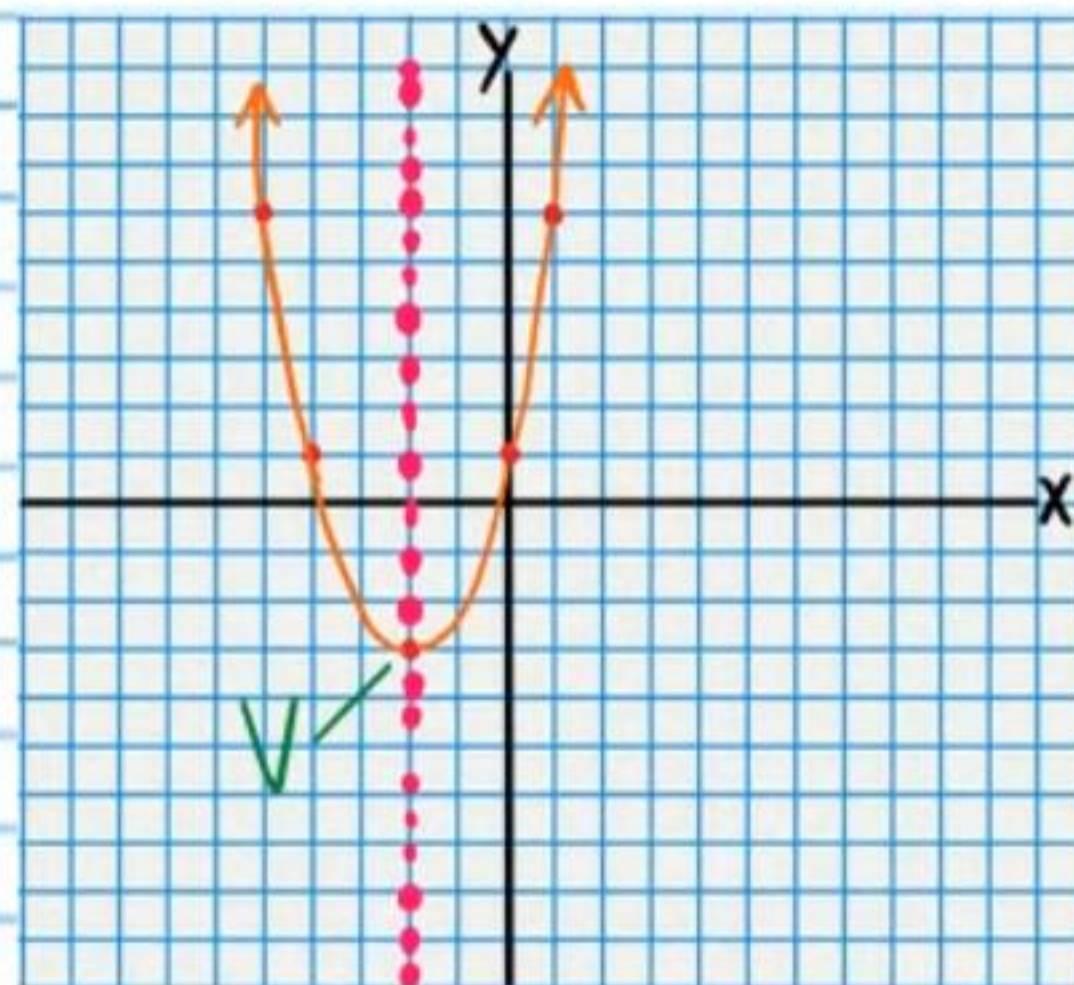
15)

$$V(-2, -3)$$

$$x = -2$$

$$(-2 + \sqrt{3}, 0) \text{ \& } (-2 - \sqrt{3}, 0)$$

$$(0, 1)$$

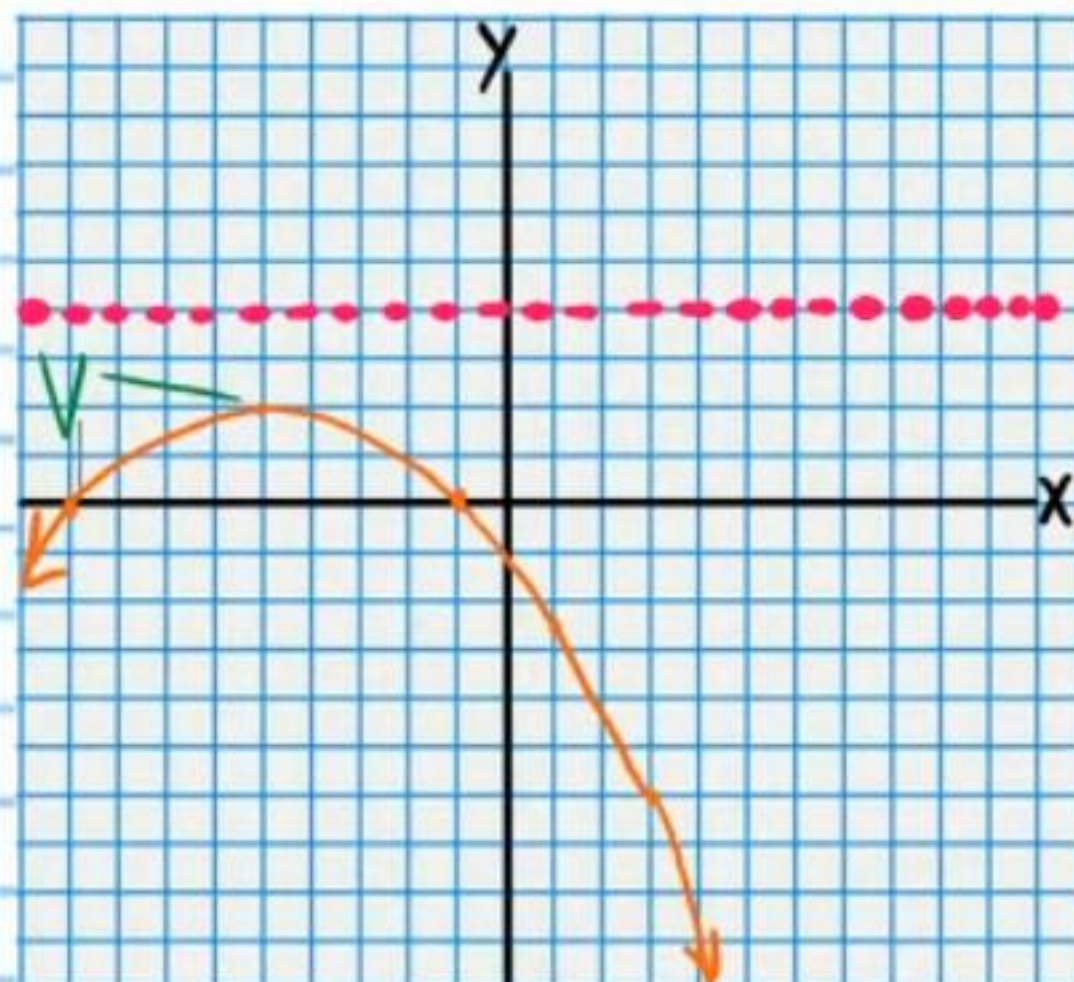


16)

$$y = 4$$

$$V(-5, 2)$$

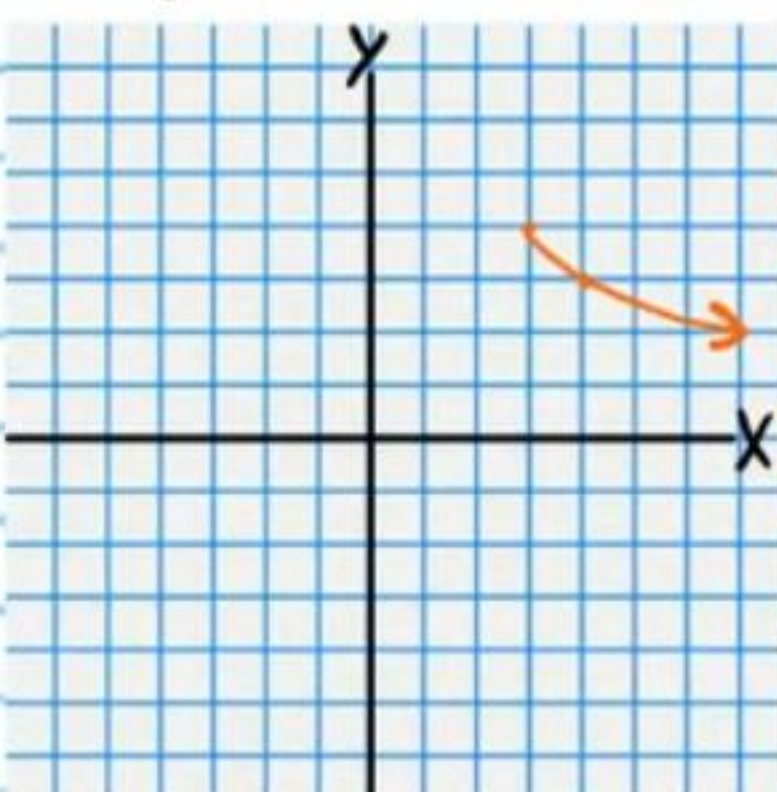
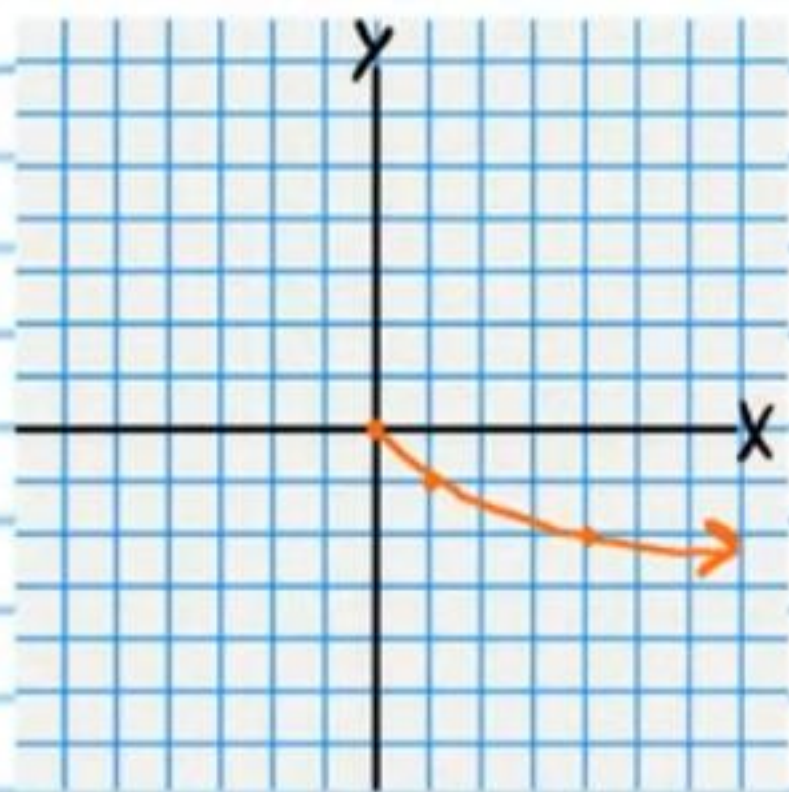
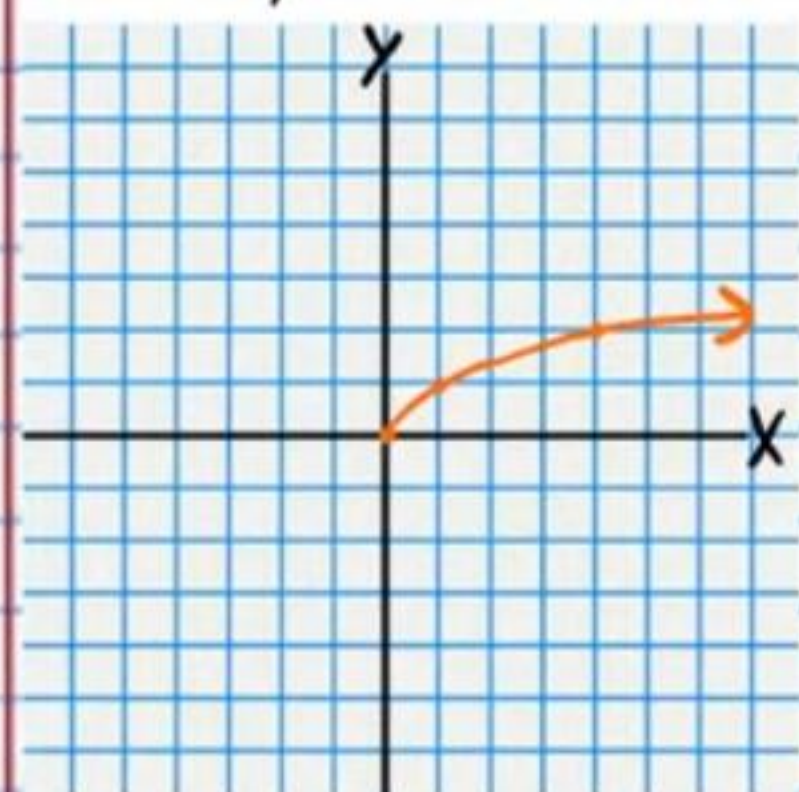
$$(-9, 0) \text{ \& } (-1, 0)$$



17) i) Parent Function  
 $y = \sqrt{x}$

Transformation #1  
 $-y = \sqrt{x}$

Final Transformation  
 $-(y - 4) = \sqrt{x - 3}$



ii) Domain:  $\{x | x \geq 3\}$

Range:  $\{y | y \leq 4\}$

18) i) Parent Function

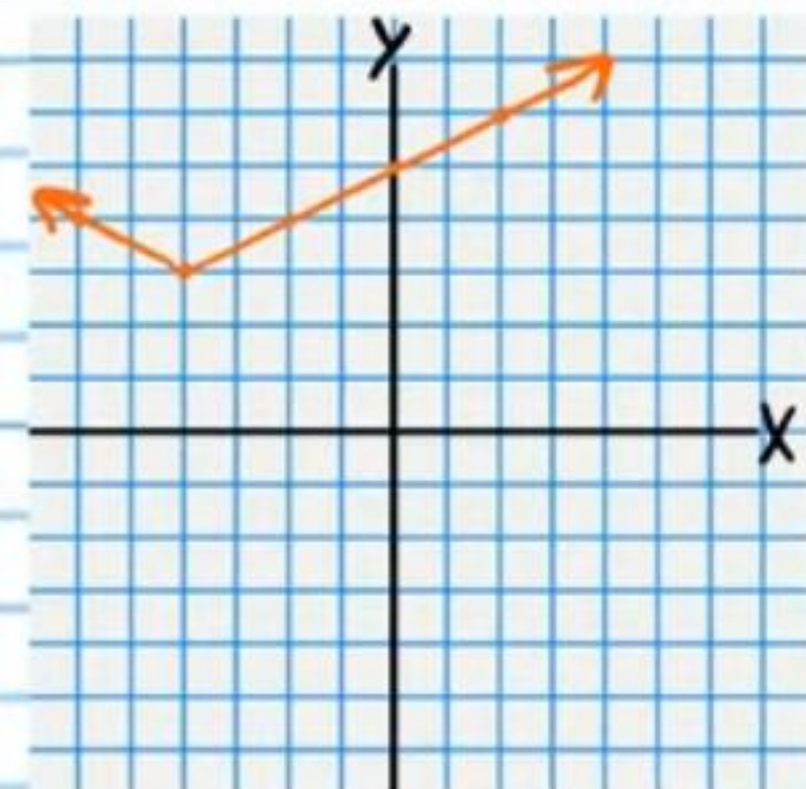
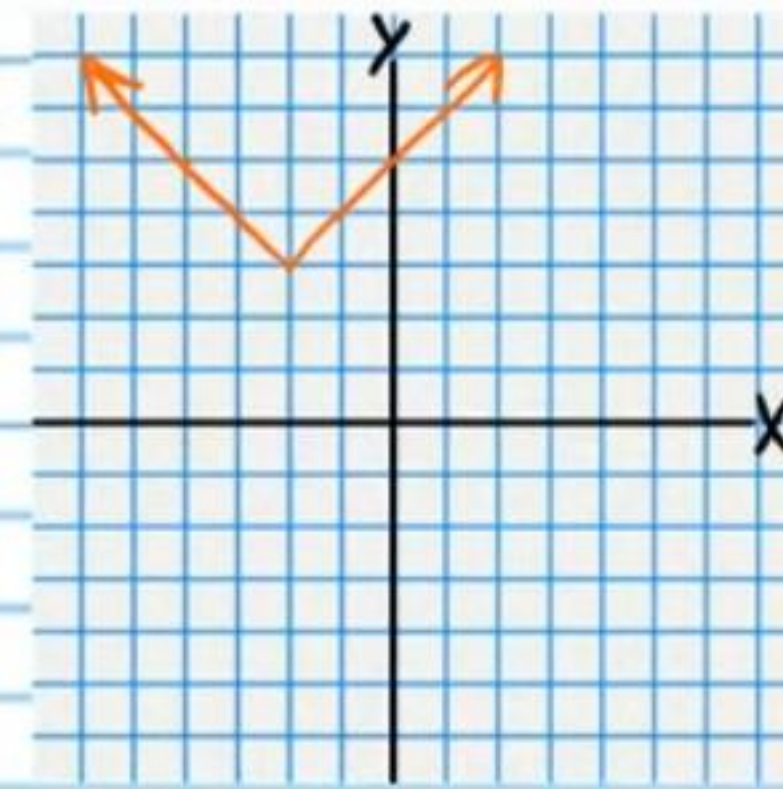
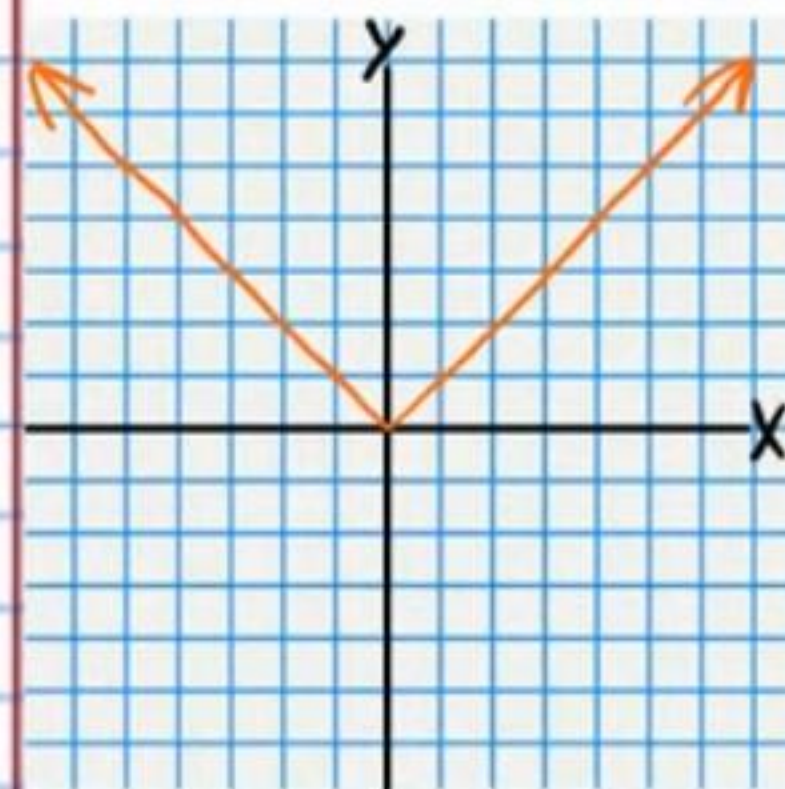
Transformation #1

Final Transformation

$$y = |x|$$

$$y - 3 = |x + 2|$$

$$y - 3 = \left|\frac{x}{2} + 2\right|$$



ii) Domain:  $(-\infty, \infty)$

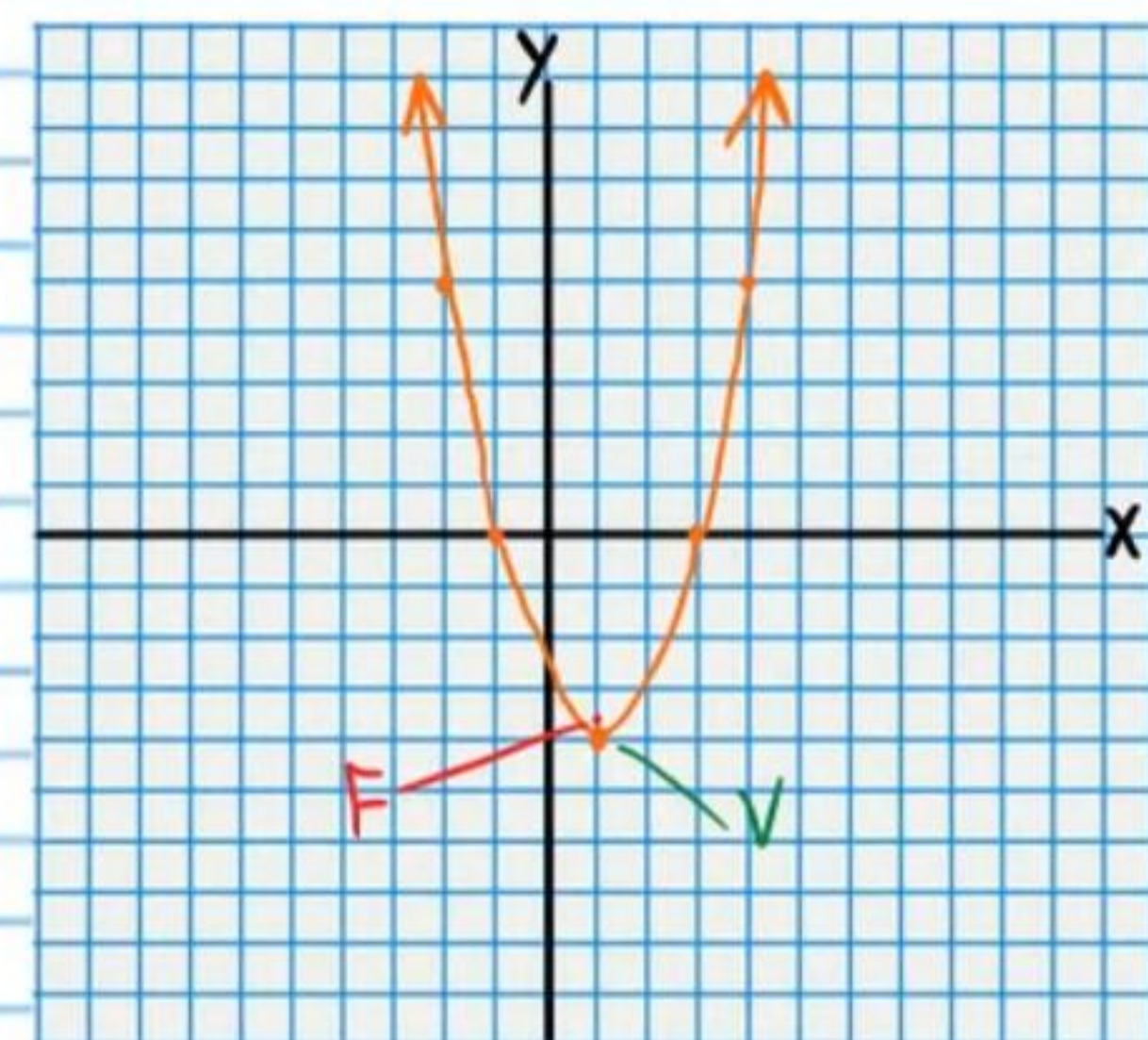
Range:  $[3, \infty)$

19)

$$F(1, -3\frac{3}{4})$$

$$V(1, -4)$$

$$(-1, 0) \text{ \& } (3, 0)$$



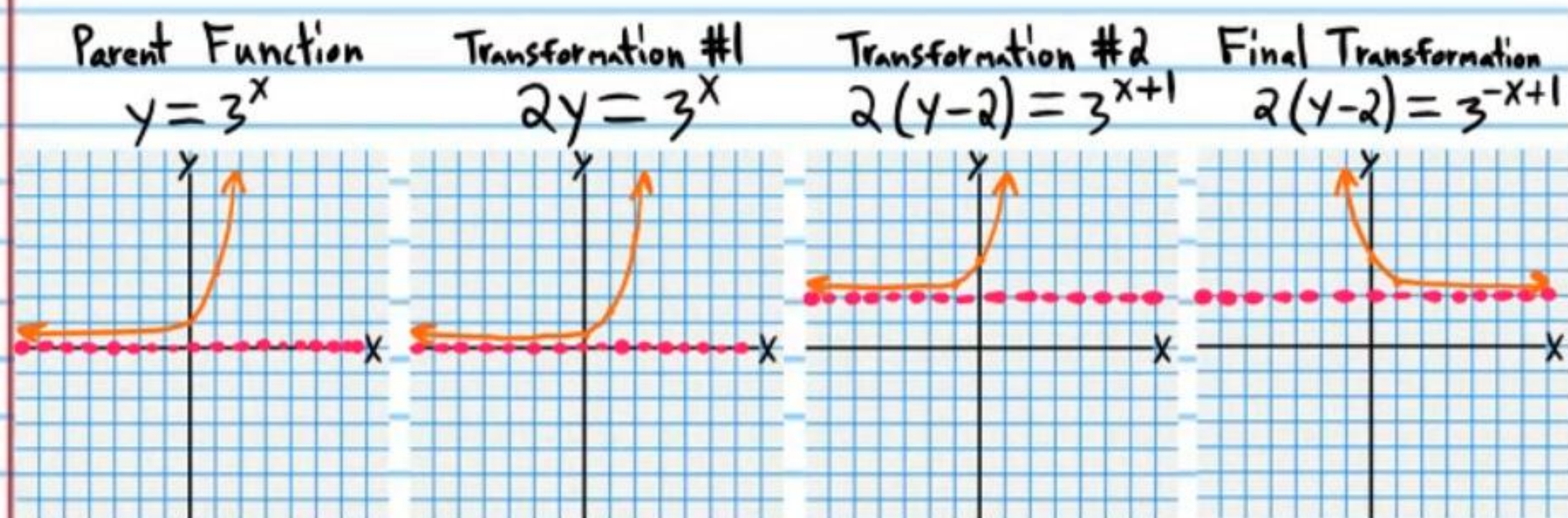
(20)  $\frac{x^3 \sqrt[4]{8x}}{4}$

(21)  $x=12$  or  $x=2$

(22) i)  $10x + 17x\sqrt{14x}$

ii)  $y=66$

(23) i)



ii) Domain: All real numbers

Range:  $\{y | y > 2\}$

(24) i) 28.29 years

ii) \$85,242.41

(25)  $(-3, 2, -3)$